

Candidate Name

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

BIOLOGY
PAPER 2

4025/2
2 hours

SPECIMEN PAPER

Additional materials:

- Electronic calculator
- Pencil (type HB recommended)
- Answer booklet

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces at the top of this page.

At the end of the examination, fasten the answer booklet securely to the question paper.

INFORMATION FOR CANDIDATES

The intended number of marks is given in brackets [] at the end of each question or part question.

This question paper consists of two compulsory Sections, **A** and **B**.

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Section A	
Section B	
TOTAL	

This specimen paper consists of 9 printed pages.

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Section A

Answer **all** questions in this section.

Write your answers in the spaces provided on the question paper.

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- 1**
- (a)** Name any **three** sub-units of carbohydrates.
1. _____
2. _____
3. _____ [3]
- (b)** State the positive colour for a reducing sugar test.
- _____
- _____ [1]
- (c)** A family is concerned that most of the foods they consume in their home contains a lot of fat.
- (i)** Write the word equation for the digestion of fats.
- _____
- _____ [2]
- (ii)** Describe the emulsion test is used to determine which of the foods contain fats.
- _____
- _____
- _____
- _____ [3]
- (d)** State any **one** use of fats in living organisms.
- _____
- _____ [1]
- [Total: 10]

2 (a) Define the term

solution,

_____ [1]

plasmolysis.

_____ [2]

- (b) **Fig. 2.1** shows an experiment that was set up to investigate a biological process. It was allowed to stand for several hours. At the beginning of the experiment, the level of the liquid in the glass tube was at **A**.

NB: Take the liquid in the beaker as distilled water.

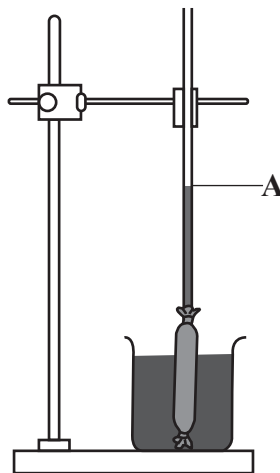


Fig. 2.1

- (i) Name the biological process under investigation.

_____ [1]

- (ii) Describe the expected changes to the levels of the liquids after several hours in the

glass tube,

beaker.

[2]

- (iii) Explain the answer in **b(ii)**.

[2]

- (iv) Suggest changes that can be made to make the level at **A** move further and faster.

[2]

[Total: 10]

- 3** (a) (i) Name the branch of biology concerned with the study of living organisms and their environment.

[1]

- (ii) Suggest any **three** ways a community may conserve the environment.

[3]

- (b) (i) Describe the reproduction of bacteria.

_____ [1]

- (ii) Outline any **three** conditions necessary for bacteria to survive.

_____ [3]

- (iii) Some food in homes may be spoiled due to the action of bacteria.

Suggest **two** ways of reducing the spoiling of food in a home.

1. _____
2. _____ [2]

[Total: 10]

- 4 (a) (i) State the cause, and any **two** effects of sickle cell anaemia.

cause _____

effects _____
_____ [3]

- (ii) Explain any **one** of the effects mentioned in a(i).

_____ [2]

- (b) (i) Define the term *gene*.

_____ [1]

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- (ii) In lions, light brown fur is dominant over white fur. A light brown male lion mated with a white female lion and produced five light brown cubs and two white cubs.

NB: Fur colour in lions is not sex linked.

Use a genetic diagram to explain the outcome of this cross.

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[4]

[Total: 10]

Section B

Answer any **three** questions from this section.

Write your answers on the separate answer booklet provided.

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- 5 (a) (i) Define the term *enzyme*. [2]
- (ii) State any **four** properties of enzymes. [4]
- (b) Explain how enzymes work to convert starch to sugars. [4]
- (c) A company uses amylase during brewing to convert starch to sugars at a temperature of 60 °C.
- (i) Predict the effect of this temperature on the enzyme's activity. [2]
- (ii) Suggest how the efficiency of the amylase may be improved. [2]
- (d) (i) Name any **two** enzymes added to a biological washing powders. [2]
- (ii) Explain the role of the enzymes named in **d(i)**. [2]
- (e) Explain why it is important to maintain other factors constant in experiments on properties of enzymes. [2]

[Total: 20]

- 6 (a) (i) Define, with reference to plants, the term *productivity*. [2]
- (ii) Explain how the productivity of plants in a garden is affected by
1. temperature,
 2. mineral salts,
 3. water.
- [6]

- (b) (i) Identify, with examples, the **two** types of plant pests. [4]
- (ii) State any **three** cultural methods of controlling plant pests. [3]
- (c) (i) Describe the function of nitrogen and phosphorus in plants. [2]
- (ii) A farmer notices that maize plants in the field have yellow leaves and brown edges. The plants also have weak stems.
- Suggest steps the farmer can take to reverse the symptoms observed in the plant. [3]

[Total: 20]

- 7 (a) (i) Describe how an image is formed in the human eye. [3]
- (ii) Name the part of the eye that
1. protects it from mechanical damages,
 2. forms the outer covering,
 3. changes the size of the pupil,
 4. forms the blind spot. [4]
- (b) (i) State any **two** eye defects. [2]
- (ii) Describe how the eye focuses on a distant object. [3]
- (c) Explain, the term *infectious disease*. [2]
- (d) With reference to typhoid
- (i) state the causative agent, [1]
 - (ii) outline the signs and symptoms, [2]
 - (iii) Suggest any **three** ways of reducing its spread. [3]

[Total: 20]

- 8 (a) (i) State any **four** characteristics of kingdom *Plantae*. [4]
- (ii) Identify any **three** threats to biodiversity. [3]
- (b) (i) Define the term *ecosystem*. [2]
- (ii) Compare properties of clay soil to sandy soil. [4]
- (c) A gardener notices that the following organisms are part of the community in the garden: aphids, locusts, ladybird beetles, spiders and birds. The garden is an example of an ecosystem.
- (i) Construct a food chain involving **three** trophic levels using any of these organisms. [3]
- (ii) Explain the effects, on the ecosystem, of removing any one of these organisms. [4]
- [Total: 20]

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